

Challenge Problems Week 3

The Clifford Group

UCQ Education Team

Problem 1. Verify the identities

$$\begin{array}{c} \text{---} \boxed{H} \text{---} \boxed{X} \text{---} \boxed{H} \text{---} = \text{---} \boxed{Z} \text{---} \\ \text{---} \boxed{H} \text{---} \boxed{Z} \text{---} \boxed{H} \text{---} = \text{---} \boxed{X} \text{---} \end{array}$$

The *Clifford group* is the set of unitary matrices made by combining gates from the set $\{H, CNOT, S\}$.

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = \sqrt{Z}$$

Any matrix in the Clifford group C satisfies the following property: for any matrix P made of tensor products of Pauli matrices $\{I, X, Y, Z\}$ and scaling by ± 1 or $\pm i$ (so e.g. $iY \in P$), $CPC^\dagger$ is also a tensor product of Pauli matrices. (We proved this for H , one of our generators.)

Problem 2. Find a one-qubit unitary gate T such that $TXT^\dagger$ is not a multiple of a Pauli matrix. Deduce that there are unitary matrices not in the Clifford group.

Special Problem (for hardcore mathematicians only). Prove that the limit of matrices in the Clifford group is in the Clifford group. Deduce that $\{H, CNOT, S\}$ is not a universal gate set.